

SOP 201: MRI Safety Screening

1 Summary

All individuals must be screened before they enter the Zone 3. This document establishes screening procedures at the Notre Dame Human Neuroimaging Center (ND-HNC).

2 Scope

These policies apply to all individuals who enter Zones 3 & 4 (B15, B17, B17a; Form RO03).

3 Definitions

- **Electronic Lock:** All doors at the ND-HNC are controlled by electronic locks. The electronic lock system recognizes the owner's Irish1Card, logs the time of the attempted entry, and unlocks the door if the owner has been granted access.
- **Lead:** Primary person in Zones 3 and 4 responsible for the Visitor's safety, in charge during emergency responses. Level 4 operator or MRI Technologist.
- **Magnet Room:** B17A; The room within the MRI suite that contains the MRI scanner. It is accessed from within the MRI Control room (B17) through a doorway that is controlled by an Electronic Lock.
- **ND-HNC:** Notre Dame Human Neuroimaging Center, Lower Level of Veldman Family Psychology Clinic.
- **Operator:** MRI Technologist or Level 4 researcher, responsible for conducting research operations and maintaining equipment. The Lead during emergency responses.
- **Visitor:** An individual who has not taken the ND-HNC MRI Safety Course but has a legitimate reason for entering the MRI suites. Research participants and their legal guardians are Visitors. Other Visitors may include: individuals providing/receiving a tour of the VFPC and ND-HNC, faculty and staff who have not received MRI Safety Course, and research assistants.

4 Notes

- Emergency response and maintenance personnel **CANNOT** enter the Magnet Room unless they have been screened and all metal objects have been removed.

- Emergency response personnel can **ONLY** enter Zone 3 without completing an MRI Safety screener if the door to the Magnet Room has been locked by key.
- ND-HNC only stores MRI Safety forms for staff and researchers. Visitor forms are maintained by their respective research teams.

5 Policies & Procedures

The purpose of MRI Safety screening is to determine whether entry into the Magnet Room poses a health or safety risk. Accordingly, all individuals entering Zones 3 & 4 must be screened according to the procedures below.

5.1 MRI Prescreen

Prescreening is conducted for Visitor's intending to enter the Magnet Room (e.g. a research participant). Before arriving at the ND-HNC, research personnel check for potential MRI contraindications with the prospective Visitor:

- Medical implants.
 - The implant name and number must be provided to the MRI technologist at least 72 hours prior to the appointment to allow for MR compatibility research. The name and number are typically available from the surgeon's office.
 - All medical implants must be reviewed and approved by the MRI technologist **before** the Visitor enters Zone 3.
 - **Note on IUDs:** North American non-metallic intrauterine devices (e.g. Mirena) are approved. Metallic (e.g. Cu) and IUDs from other regions must be approved by the MRI Technologist.
- Other metal embedded in the body (e.g. shrapnel, unremovable piercings).
- Pregnancy.
- Claustrophobia (the Mock can be used to assess comfort levels).
- Tattoo on face or neck, including permanent makeup.

5.2 MRI Preparation

Visitors intending to enter the Magnet Room (e.g. a research participant) should be informed of the following restrictions, before arriving at the ND-HNC, to guide their preparations:

- Clothing should be free of metallic objects and clasps; bras should not have an underwire. Anti-odorant fabrics (e.g. Lululemon's Silverescent) are not MRI compatible.
- Jewelry will need to be removed, including piercings if they are metallic.
- Cosmetics such as colored contacts, magnetic lashes, and makeup or nail polish with glitter will have to be removed. Tanning lotions are not MRI compatible.
- Hair must be free of all accessories (metallic clips, barrettes, scrunchies, etc.).
- If Visitors have corrected vision, they should wear contacts or know their prescription.

5.3 MRI Safety Screen

All who enter Zone 3, regardless of whether they will enter Zone 4 (Magnet Room) fill out Form RO01: MRI Safety Screen. ND-HNC staff and research personnel submit a form (a) annually, and (b) whenever a change occurs in their MRI compatibility. Visitors fill out the form in Zone 1, and researchers with at least Level 2 access should be available to answer any questions.

- For Visitors, Form RO01: MRI Safety Screen must be filled out on the same day as entering Zone 3.
- Each question/prompt must be answered individually.
- In the case of minors, their legal guardian must fill out Form RO01: MRI Safety Screen.
 - Note: verify guardianship, which may differ from the individual who brought the minor to the center.
- The Visitor signs and dates Form RO01: MRI Safety Screen in the designated location.

5.4 MRI Safety Screen, Review 1

A researcher with at least Level 2 access will review the Visitor's responses on Form RO01: MRI Safety Screen.

- Discuss any potential contraindications, points of confusion, or concerns. Inform the Operator about any items discussed or addressed.
 - All medical implants and potentially contraindicated, unremovable items must be reviewed and approved by the MRI technologist **before** the Visitor enters Zone 3.
- After reviewing Form RO01: MRI Safety Screen, the researcher prints and signs their name in the designated location.

- If the researcher determines the Visitor is MRI compatible, they may escort the Visitor into Zones 2 & 3.

5.5 MRI Safety Screen, Review 2

A second review is required for Visitors entering Zone 4 (Magnet Room).

- The Operator conducts the second review, who must not be the same person that conducted Review 1.
- Any potential contraindications, points of confusion, or concerns are addressed, including those raised by the first reviewer.
 - All medical implants and potentially contraindicated, unremovable items must be reviewed and approved by the MRI technologist **before** the Visitor enters Zone 3.
- The Operator signs and dates Form RO01: MRI Safety Screen.
- The Visitor may enter Zone 4 only after the Operator has determined that it is safe to do so.

6 Figures

None